

Prabhaav Pillai

Computer Science Graduate Student & Software Engineer

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Education

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|-----------|--|---------------------|
| MS | Michigan State University , Computer Science | East Lansing, MI |
| | <ul style="list-style-type: none">GPA: 4.00/4.00Currently pursuing M.S. Accelerated Computer Science Program (Completed 9 graduate credits during B.S. via MSU Shared Enrollment program) | Jan 2026 – Dec 2026 |
| BS | Michigan State University , Computer Science | East Lansing, MI |
| | <ul style="list-style-type: none">GPA: 3.89/4.00 Honors College Minor in MathematicsCompleted 4-year B.S. degree in 2.5 years | Aug 2023 – Dec 2025 |

Experience

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| Extern , AI Document Intelligence Extern (Pfizer Program) | Remote |
| <ul style="list-style-type: none">Building an AI document intelligence system to automate data extraction, classification, and compliance risk detection across pharmaceutical records.Developing Python pipelines utilizing Layout-Aware OCR and RAG to query and extract structured insights from complex vendor files.Implementing document classification, dynamic file routing, and open-source LLM workflows integrated with a Gradio chatbot interface. | July 2026 – present
2 months |
| United Wholesale Mortgage , Software Engineering Capstone Developer (Click for Info: capstone.cse.msu.edu) | Pontiac, MI |
| <ul style="list-style-type: none">Developed full-stack IT Goals Dashboard using React and ASP.NET Core to visualize real-time KPIs for 2,000+ IT professionals.Built EF Core migrations, backend metric aggregation logic, dynamic gauge widgets, and sparkline UI visuals exceeding SonarQube quality standards.Collaborated with product owners to refine yearly/monthly metrics, improving operational transparency across IT teams. | Aug 2025 – Dec 2025
5 months |
| Microsoft TEALS Program , Volunteer CS Teaching Assistant | Detroit, MI |
| <ul style="list-style-type: none">Instructed 50+ students across two AP Computer Science classrooms in computational thinking, JavaScript, and algorithmic problem-solving.Coordinated with lead teachers to optimize lesson delivery, facilitate hands-on coding labs, and assist high school seniors with college readiness. | Aug 2023 – June 2025
1 year 11 months |

Projects

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|--|---------------------|
| Automotive Edge AI System - StarPilot Developer (Click for Info: https://prabhaav.me) | July 2026 – present |
| Invited to core dev team for a safety-critical open-source vehicle platform | |
| <ul style="list-style-type: none">Appointed to Development Team with full repository write access to help steer project roadmap and backend architecture.Authored and merged +1.7k lines of C++/Python into core production, deploying a real-time blind-spot monitoring system to thousands of active vehicles.Trained custom YOLO26n ONNX models on 27+ hours of crowdsourced driving video using semi-automated CAN-bus radar logging for ground truth.Engineered real-time edge performance on Snapdragon SoCs using zero-copy NV12 buffer slicing and CPU core isolation under 1.2x throttle overhead.Drove widespread community adoption (35K+ views, 175+ upvotes) and maintained production lifecycle across post-release bug fixes and dataset updates. | |

Certifications

- Certified Cloud Practitioner (Score: 981/1000) - Amazon Web Services (July 2026)

Selected Honors

- First Place Award, Education Category - MSU University Undergraduate Research Forum (UURAF, April 2024)
- Researched academic impacts of Large Language Models (LLMs) on student performance using mixed quantitative and qualitative analysis.

Skills

Languages: Python, C++, C#, JavaScript, Java, SQL

AI / Machine Learning: PyTorch, YOLOv8/26, Computer Vision, RAG, Layout OCR, ONNX Runtime, Model Optimization

Edge AI & Systems: Embedded AI, Real-time Inference, Linux, CMake, Performance Profiling, Networking

Frameworks & Tools: React, ASP.NET Core, Docker, REST APIs, Git, GitHub, Swagger UI, Fast API, Jupyter Notebook

Cloud & Databases: AWS (Certified Cloud Practitioner), Azure SQL, Entity Framework Core